## 4.6 Querying models in views

Many of the views you want to write need to be parameterized. You wouldn't write a view for every musician in the database; instead, you want to write a single view that takes a musician as an argument. There are two ways to parameterize a view:

- URLs based arguments
- Query parameters

Figure 4.5 shows both types of parameterized URLs.

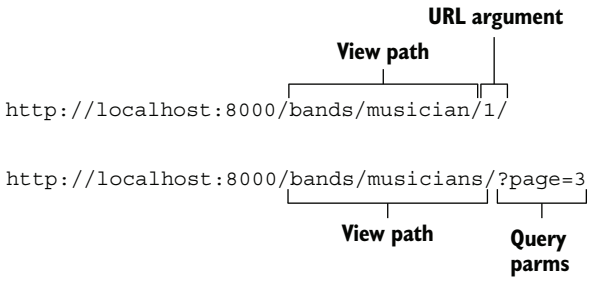



**Figure 4.5 URL-based arguments and query parameters**

You may be more familiar with query parameters if you've ever seen key–value pairs in a URL after a question mark. Django supports these but also allows you to capture parts of a URL as arguments to your view. Captured parts is the recommended way of using arguments with your URL, but query parameters still get used for optional behaviors in a view.

### 4.6.1 Using URL arguments

Let's start with Django's preferred case: URL based arguments. To build a page that displays information for a single musician, you need a way of specifying which musician. As your database already has a unique identifier for each musician, the typical solution to this problem is to use the database ID as the argument to the view.

A view to show musician details uses the ID of a `Musician` object as an argument. Remember that the URL `path` object is what defines the mapping between the URL in the browser and the called view. To support arguments to the view, you specify this in the `path` object declaration, indicating the arguments and their type using angle brackets. Inside the view, you perform a look-up of the `Musician` object for the given ID from the database. The resulting object gets rendered through a template.

Django provides a shortcut named `get_object_or_404()` to help you fetch an object or error out if the argument is invalid. This shortcut takes two or more arguments. The first argument is the `Model` object to query—in this case, the `Musician`. The second and subsequent arguments get used to form the look-up query. They use the same format as the query manager's `.get()` method, as that's what the shortcut is